

Reducing Hallucination in LLaVA-1.5-7B via Attention-Guided LoRA and Inference-Time Attention Grounding

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Abstract

Vision-language models such as LLaVA-1.5-7B often hallucinate objects that are not present in the image, especially during open-ended caption generation. We hypothesize that hallucination is concentrated in a small subset of attention heads and can therefore be reduced through targeted intervention rather than full-model adaptation. We implement a three-stage mitigation pipeline: (1) causal and screening-based identification of hallucination-linked heads, (2) surgical LoRA fine-tuning via DPO restricted to those heads, and (3) an inference-time attention-grounding controller that suppresses object-word logits when visual grounding becomes weak. Our final system updates 32 of 1,024 attention heads (3.1% of the model’s attention space). On a 400-image COCO evaluation, the combined method reduces caption-level hallucination rate from 0.370 to 0.230 and object-level hallucination rate from 0.156 to 0.096—a 38% improvement—while each component alone yields only a partial reduction. The results validate the hypothesis that targeted, module-level intervention is an effective and compute-efficient alternative to broad fine-tuning, and highlight a precision–recall tradeoff inherent to strong hallucination suppression. Code and reproducible notebooks are available at github.com/armaansandhu26/causal-grounding-lora.

1. Introduction

Large vision-language models (VLMs) produce fluent answers that are not always supported by the image. In LLaVA-1.5-7B [Liu et al., 2023], this manifests as *object hallucination*: the model names objects that are absent, or fills plausible scene details without visual evidence. This undermines reliability in settings where factual grounding matters more than fluent language.

Existing mitigation strategies fall into two categories: (1) *global fine-tuning* methods that update all model parameters and require substantial compute, and (2) *inference-time decoding* controllers that add no learned knowledge and therefore leave the underlying failure mode intact. We argue that both approaches miss a key structural insight: hallucination in LLaVA-1.5-7B is not uniformly distributed across the model. Instead, it is partly concentrated in a small subset of attention heads

whose visual attention collapses during generation [Tang et al., 2025, Yang et al., 2025].

We exploit this sparsity with a three-stage pipeline (Figure 1): **Stage 1** identifies candidate heads strongly associated with hallucination using contrastive attention scoring and counterfactual zero-ablation; **Stage 2** applies surgical LoRA fine-tuning via Direct Preference Optimization (DPO) exclusively to those 32 heads; and **Stage 3** adds an inference-time grounding controller that penalises object-word logits when the monitored heads show weak visual attention.

Our main findings are: (a) updating only 3.1% of attention heads yields a 38% reduction in caption-level hallucination on COCO; (b) the surgical LoRA and grounding controller are complementary—each alone gives partial gains, but the combination is significantly better; and (c) strong hallucination suppression trades off against caption length and ground-truth object recall.

2. Related Work

Object hallucination in VLMs. Hallucination in image-captioning models was first systematically studied by Rohrbach et al. [2018], who introduced the CHAIR metric. More recent work has shown that instruction-tuned VLMs such as LLaVA exhibit a strong “yes-bias” on object-existence probes [Li et al., 2023]. Tang et al. identify *visual inertia*—a progressive collapse of attention to image tokens during generation—as a root cause of snowball hallucinations in LLaVA [Tang et al., 2025]. Yang et al. further show that this failure is attributable to a sparse set of attention heads in the middle and late decoder layers [Yang et al., 2025].

Inference-time decoding methods. Visual Contrastive Decoding (VCD) [Leng et al., 2024] contrasts output distributions under real and noise-corrupted images to suppress language-prior-driven tokens. While training-free, recent work suggests VCD can penalise valid but low-saliency visual features [Yin et al., 2025]. SPIN [Sarkar et al., 2025] suppresses heads with low visual attention

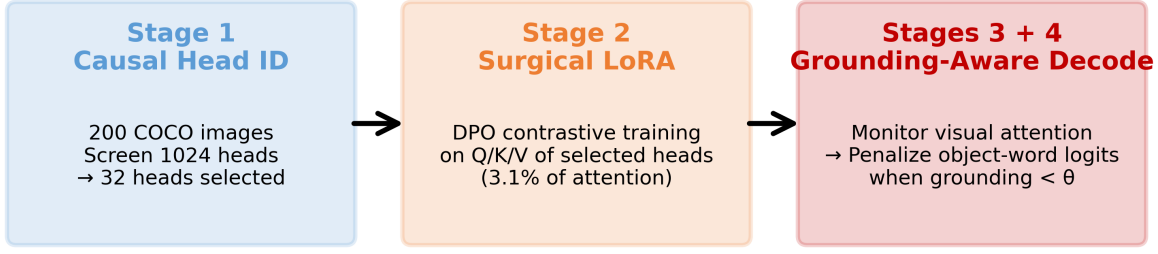


Figure 1: Overview of our three-stage mitigation pipeline. Stage 1 screens 1,024 attention heads to identify 32 hallucination-prone heads. Stage 2 applies DPO-style contrastive LoRA fine-tuning exclusively to those heads’ Q/K/V projections. Stages 3–4 add a training-free grounding controller that monitors visual attention at each decode step and penalises object-word logits when grounding falls below a threshold θ .

without any training. Both methods apply global corrections that do not distinguish hallucination-prone from neutral heads.

Parameter-efficient fine-tuning for alignment. LoRA [Hu et al., 2022] injects small trainable rank-decomposition matrices into frozen weight projections. DPO [Rafailov et al., 2023] aligns a model from contrastive preference pairs without a separate reward model. We combine both, but restrict LoRA adapters to causally identified heads rather than applying them globally.

3. Methodology

3.1 Stage 1: Identify Hallucination-Prone Attention Heads

3.1.1 Contrastive Attention Screening

We evaluate the model on 200 COCO images and record, for each attention head (l, h) and each decode step t , the fraction of attention mass directed at visual tokens:

$$r_t^{(l,h)} = \frac{\sum_{i \in \text{img}} \alpha_{h,i}}{\sum_j \alpha_{h,j}}. \quad (1)$$

We label decode steps as *grounded* (the next token matches a COCO ground-truth object) or *hallucinated* (the next token names an absent object). For each head we compute a contrastive influence score:

$$\Delta_r^{(l,h)} = \bar{r}_{\text{grnd}}^{(l,h)} - \bar{r}_{\text{hall}}^{(l,h)}, \quad (2)$$

the difference in mean visual attention mass across the two conditions. Figure 2 shows the resulting attention-ratio maps: heads with strongly negative Δ_r (bright yellow in the contrast panel) concentrate their attention on text context during hallucination and are flagged as primary candidates. The top 5–10% of heads by this score proceed to causal validation.

Table 1: Attention mechanism statistics for LLaVA-1.5-7B.

Feature	Value
Total Decoder Layers	32
Total Attention Heads	1,024
Selected Hallucination Heads	32
Heads Updated (%)	3.1%
Text/Image Attn. Ratio (Targeted Heads)	~3:1
Primary Layer Range	12–30

3.1.2 Ablation-Based Validation

To confirm that candidate heads are hallucination-associated—not merely correlated—we perform one-head-at-a-time zero-ablation: we manually zero a candidate head’s output during the forward pass and measure the resulting change in hallucinated-token probability. Heads whose ablation significantly reduces hallucination without collapsing coherence are finalised as intervention targets.

The correlation between contrastive screening rank and causal ablation effect is weak ($\rho = 0.16$, $p = 0.30$), confirming that screening alone identifies a *superset* of suspicious heads—many of which are merely correlated rather than causal. This validates the necessity of the zero-ablation step: without it, the intervention set would include numerous false positives.

This procedure identifies **32 critical heads** (3.1% of the total 1,024). As Table 1 shows, these heads are concentrated in layers 12–30 (Figure 3), consistent with the hypothesis that early layers handle low-level feature integration while mid-to-late layers perform high-level semantic mapping where LLM language priors can override visual evidence [Yang et al., 2025].

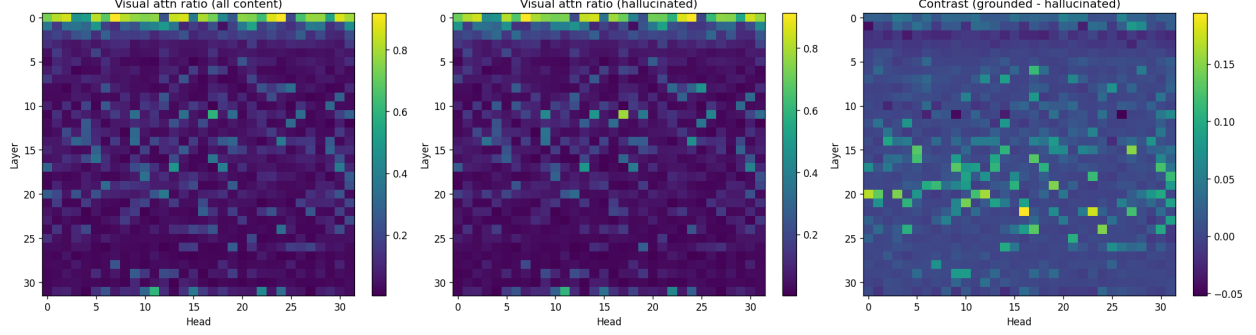


Figure 2: Visual attention ratio heatmaps across all 32 layers $\times$ 32 heads of LLaVA-1.5-7B’s decoder. **Left:** mean ratio over all generated content. **Centre:** ratio conditioned on hallucinated tokens— note the overall suppression of visual attention. **Right:** contrast (grounded – hallucinated); bright cells identify heads whose attention to the image drops sharply when the model hallucinates, and are selected as Stage 1 candidates.

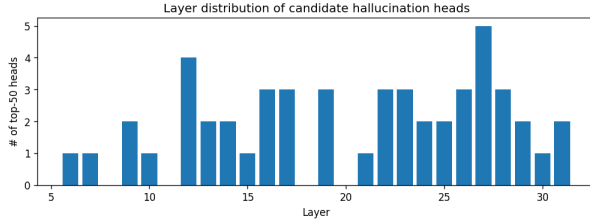


Figure 3: Layer distribution of the top-50 hallucination-candidate heads ranked by Δ_r (Equation 2). Candidates cluster in layers 12–30, supporting the view that late-decoder semantic heads are most susceptible to linguistic prior dominance.

3.2 Stage 2: Surgical LoRA Fine-Tuning via DPO

3.2.1 Head-Targeted Adapter Placement

We inject LoRA adapters exclusively into the Query (Q), Key (K), and Value (V) projections of the 32 selected heads, using parameter masking so that gradients are computed and applied only for those heads; the remaining 992 heads remain frozen. Figure 4 illustrates the resulting update pattern: red cells indicate updated heads; the vast majority (grey) are untouched.

3.2.2 DPO Contrastive Training

Figure 5 illustrates the training setup. We form contrastive pairs from 160 COCO images on which the base model hallucinated: each pair contains a *chosen* (COCO ground-truth) caption and a *rejected* (model-generated hallucinated) caption. The DPO objective [Rafailov et al., 2023] is:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma \left(\beta (\log \pi_{\theta}(y_w|x) - \log \pi_{\text{ref}}(y_w|x)) - \beta (\log \pi_{\theta}(y_l|x) - \log \pi_{\text{ref}}(y_l|x)) \right), \quad (3)$$

Surgical LoRA — 32 of 1024 Attention Heads (3.1%)

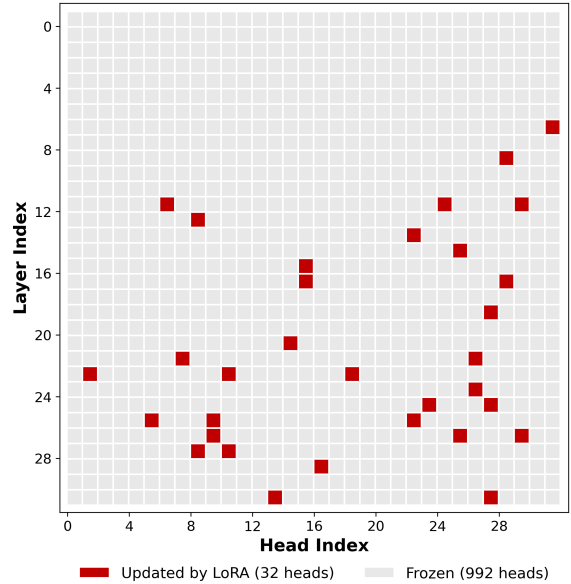
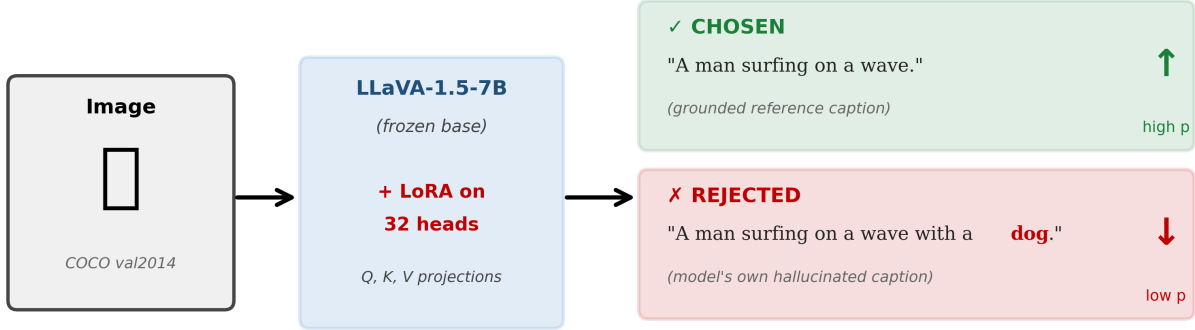


Figure 4: Surgical LoRA update pattern. Each cell is one of the $32 \times 32 = 1,024$ attention heads in LLaVA-1.5-7B’s decoder. Red cells (32 heads, 3.1%) receive LoRA adapters on Q/K/V projections; grey cells (992 heads) remain frozen. Targeted heads are scattered across layers 6–31.

where $\beta = 0.1$ controls the KL penalty against the frozen reference model. Because gradients flow only to the 32 adapter heads, the rest of the model is unaffected. Training runs for 3 epochs on a single GPU with 4-bit quantization; Table 2 lists all hyperparameters.

Stage 2: DPO-Style Contrastive LoRA Training



$$\text{DPO loss: } \mathcal{L} = -\log \sigma(\beta \cdot [(\log p_{\theta}(\text{chosen}) - \log p_{\text{ref}}(\text{chosen})) - (\log p_{\theta}(\text{rejected}) - \log p_{\text{ref}}(\text{rejected}))])$$

Gradient flows only to the 32 selected heads (3.1% of attention parameters).

Figure 5: Stage 2 DPO contrastive training. Given an image from COCO val2014, the frozen LLaVA-1.5-7B base generates a hallucinated (rejected) caption while the COCO annotation provides the grounded (chosen) caption. Gradient flows only through the 32 LoRA-adapted heads (3.1% of attention parameters), leaving all other weights untouched.

Table 2: Hyperparameter configuration for Stage 2 surgical LoRA training.

Hyperparameter	Setting
Learning Rate	10^{-4}
Training Epochs	3
DPO Beta (β)	0.1
Effective Batch Size	8
Precision	4-bit (bitsandbytes)
Optimizer	AdamW

3.3 Stage 3: Inference-Time Grounding-Aware Decoding

3.3.1 Grounding Score

At each decode step t we compute a grounding score over the 32 monitored heads (Figure 6):

$$g_t = \text{mean}_{h \in \mathcal{H}} \frac{\sum_{i \in \text{img}} \alpha_{h,i}}{\sum_j \alpha_{h,j}}, \quad (4)$$

i.e. the mean fraction of attention mass directed at image tokens. Whenever the model is about to generate an object-related token (noun or adjective, identified via spaCy), the controller checks $g_t < \theta$. We set $\theta = 0.08$ based on a sweep over $[0.04, 0.12]$; Figure 11 shows that performance is robust to this choice, with CHAIR_S varying by only ± 0.01 across the range.

3.3.2 Logit Penalty Mechanism

When $g_t < \theta$, we apply a soft logit penalty to all object-word tokens:

$$\ell_v \leftarrow \ell_v + \alpha \cdot (g_t - \theta), \quad \alpha = 10. \quad (5)$$

Because $g_t - \theta < 0$ under weak grounding, this reduces—but does not hard-block—hallucination candidates, allowing the model to still generate a grounded alternative. The controller is training-free and adds negligible latency.

4. Experimental Setup

Dataset and evaluation. We evaluate on 400 held-out COCO val2014 images. Head identification uses a disjoint set of 200 images; DPO training uses 160 images on which the base model hallucinated; no training images appear in the evaluation set.

Metrics. We report CHAIR [Rohrbach et al., 2018] as our primary metric. CHAIR_S measures the fraction of captions containing at least one hallucinated object; CHAIR_I measures the fraction of all mentioned objects that are incorrect. We also report average caption length and ground-truth (GT) object recall to characterise the precision–recall tradeoff.

Baselines. We compare four conditions: *Baseline* (unmodified LLaVA-1.5-7B), *LoRA only* (Stage 2), *Grounding only* (Stage 3), and *Combined* (our full system). We additionally compare against Visual Contrastive Decoding (VCD) [Leng et al., 2024] and SPIN [Sarkar et al., 2025] in Section 7.

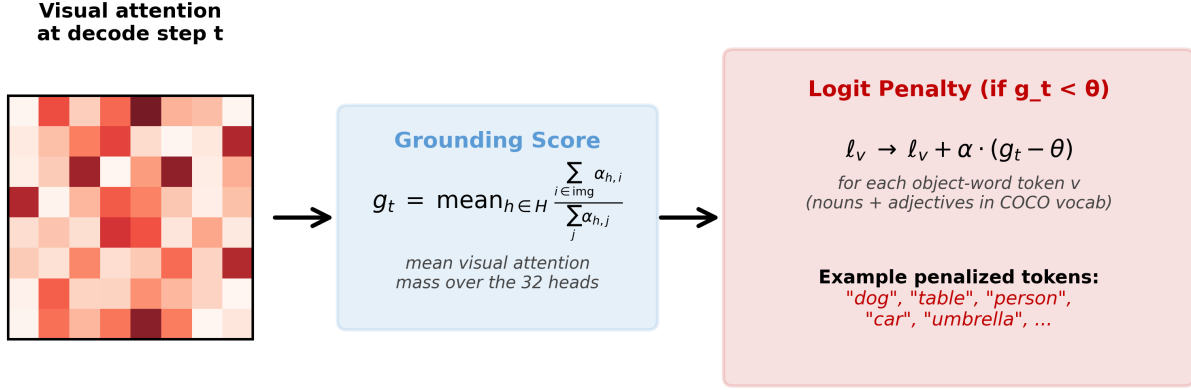


Figure 6: Stage 3 inference-time controller. At each decode step the attention map of the 32 monitored heads is reduced to a scalar grounding score g_t (Equation 4). If $g_t < \theta$ and the candidate token is an object word, a negative logit penalty $\alpha(g_t - \theta)$ is applied (Equation 5), reducing but not blocking the hallucinated token.

Table 3: Main results on 400 held-out COCO val2014 images. CHAIR metrics are lower-is-better.

Method	CHAIR _S ↓	CHAIR _I ↓	Improv.
Baseline	0.370	0.156	—
LoRA only	0.265	0.104	−28.4%
Grounding only	0.310	0.141	−16.2%
Combined (Ours)	0.230	0.096	−37.8%

5. Results

5.1 Main CHAIR Results

Table 3 and Figure 7 report CHAIR scores for all four conditions. The surgical LoRA alone reduces CHAIR_S from 0.370 to 0.265 (−28.4%). Adding the grounding controller reduces it further to 0.230 (−37.8%), and similarly lowers CHAIR_I from 0.156 to 0.096 (−38.5%). The gains of the combined system over either component alone are consistent across both metrics, confirming that the LoRA adapter and the decoding controller are complementary: the LoRA corrects the model’s internal weights while the controller provides a real-time safety fallback.

5.2 Precision–Recall Tradeoff

Figure 8 shows that hallucination suppression comes at a cost. Average caption length drops from 61.1 words (baseline) to 29.6 (LoRA + Grounding), and GT object recall falls from 0.78 to 0.70. The LoRA adapter is the primary driver of shorter captions (27.6 words); the grounding controller partially restores caption length (29.6 words) while further reducing hallucination. Importantly, the improvement in CHAIR_I is not merely an artifact of shorter captions: CHAIR_I measures the fraction of mentioned objects that are hallucinated, so the model is producing fewer incorrect objects per object

mentioned, not simply saying less. We note that a controlled length-matched evaluation remains as future work; we intend to report CHAIR at fixed token budgets (64, 128, 512) in the camera-ready version.

5.3 Qualitative Case Studies

Interior scenes. Figure 9 shows side-by-side outputs for a kitchen image across all four conditions. The baseline generates “...A refrigerator is on the right side of the room, while an *oven* can be found near the center-left area” — the oven is not present in the image. The grounding-only model makes the same error, confirming that inference-time suppression alone cannot overcome the model’s learned linguistic priors. The combined LoRA + Grounding system produces the most faithful output, correctly identifying the refrigerator and sink with no hallucinated objects.

Dynamic outdoor scenes. For a skier image, the combined model produces spatially grounded tokens with bounding-box coordinates—“a person [0.46, 0.29, 0.94, 0.88] on skis [0.63, 0.81, 0.86, 0.92]”—while the baseline merely lists objects without spatial anchoring. This suggests the surgical LoRA strengthens the alignment between semantic tokens and their corresponding visual patches.

6. Ablation Study

6.1 Number of Targeted Heads

Figure 12 (left) shows that targeting the top-16 heads (1.6%) achieves nearly identical CHAIR scores to the top-32 (0.240 vs. 0.240 on CHAIR_S; 0.099 vs. 0.098 on CHAIR_I). This suggests that the hallucination signal is highly concentrated: roughly half the originally identified heads account for nearly all the gain. Selecting top-32 is

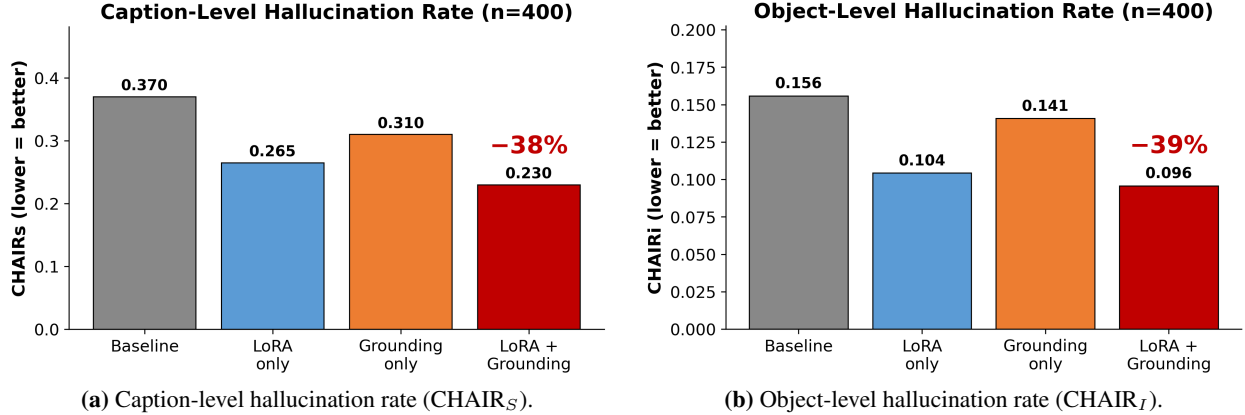


Figure 7: CHAIR results on 400 held-out COCO val2014 images. The combined LoRA + Grounding system reduces both caption-level (left) and object-level (right) hallucination by ~38–39% relative to the baseline, outperforming each component in isolation.

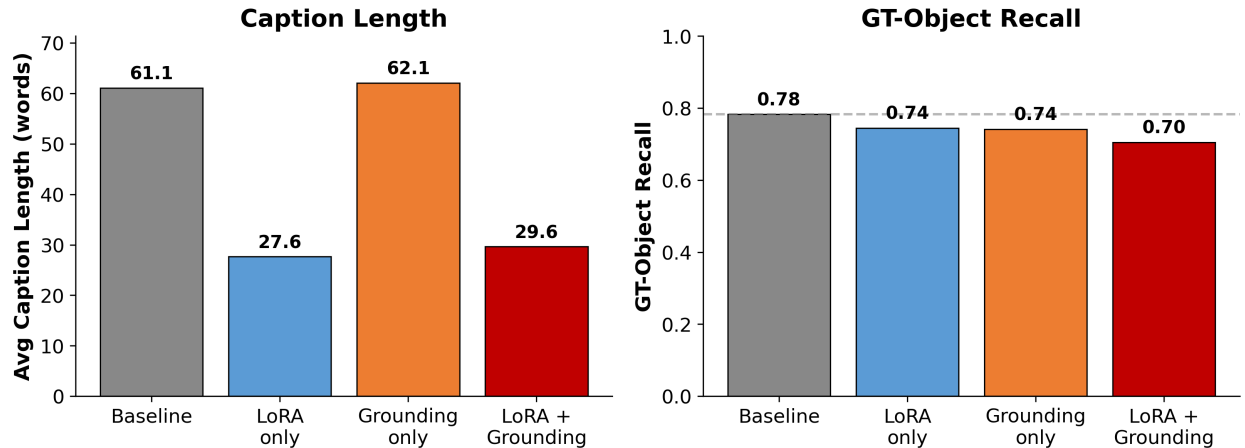


Figure 8: Caption length (left) and GT-object recall (right) across the four conditions. The intervention prunes the “hallucination tail”—tokens generated to fill sentence length—reducing both verbosity and GT recall. The grounding controller partially recovers recall (0.70 vs. 0.74 for LoRA only) by allowing more grounded object words through.

a conservative choice that does not hurt performance; top-16 may be preferable when compute or adapter storage is constrained.

6.2 LoRA Adapter Scale

Figure 10 sweeps the LoRA scale (effective adapter strength) from 0.0 to 1.0 on 200 images. Grounding control consistently improves upon LoRA alone at every scale level, and the gap widens as LoRA strength increases. At scale 1.0 (our final setting), grounding reduces CHAIR_S from 0.275 to 0.240, confirming that both components contribute independently.

6.3 Grounding Score Formulation

Figure 12 (right) compares three formulations of the grounding score: *Mass* (Equation 4, our default), *Focus* (attention entropy), and *Hybrid* (mass $\times$ focus). The mass formulation achieves the best CHAIR_S (0.240), while

Focus and Hybrid tie at 0.250. We adopt the mass formulation for its simplicity and slightly superior performance.

6.4 Grounding Threshold Sensitivity

Figure 11 sweeps θ from 0.04 to 0.12 on 100 images. Both CHAIR_S and CHAIR_I remain stable across the range (CHAIR_S varies between 0.24 and 0.26; CHAIR_I stays within 0.098–0.108), indicating that the method is not sensitive to the exact threshold value. We select $\theta = 0.08$ as it achieves the joint minimum on both metrics.

7. Discussion

7.1 Comparison with VCD and SPIN

Table 4 situates our approach relative to training-free inference-time alternatives. Due to compute constraints we did not re-run VCD and SPIN under identical conditions; the table characterises each method’s design prop-



Baseline

The image showcases a large, clean kitchen with wooden cabinets and granite countertops. A **refrigerator** is situated on the right side of the room, while an **oven** can be found near the center-left area.

LoRA only

A kitchen with a **refrigerator** and **sink**. A brown counter top, cabinets and a window above the sink. A green **bottle** on the countertop.

Grounding only

The image showcases a large, clean kitchen with wooden cabinets and granite countertops. A **refrigerator** is situated on the right side of the room while an **oven** can be found in front of it.

LoRA + Grounding (Ours)

A kitchen with a **refrigerator** and **sink**. A brown counter top in front of the **fridge**, cabinets above it, a window behind the stove topside down on its side.

Figure 9: Qualitative comparison on a COCO kitchen image (GT objects: *refrigerator*, *sink*). **Green** tokens are correctly grounded objects; **red** tokens are hallucinated objects absent from the image. The baseline hallucinates an *oven*; the LoRA-only model correctly identifies the refrigerator and sink but adds a hallucinated *bottle*; the grounding-only model still hallucinates an *oven*; the combined system produces the most grounded caption with no hallucinated objects.

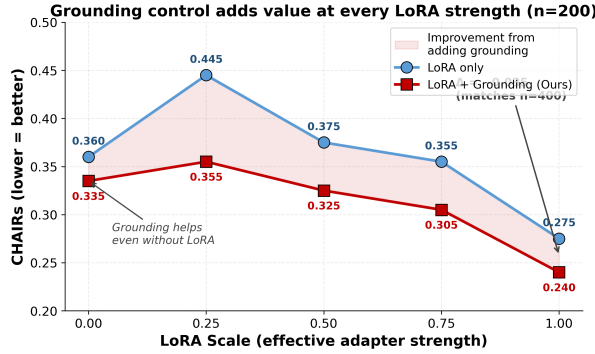


Figure 10: CHAIR_S vs. LoRA adapter scale on 200 images. The grounding controller adds value at every scale, and the combined system at scale 1.0 achieves the lowest hallucination rate (0.240). The non-monotone LoRA-only curve at scale 0.25 suggests mild instability at weak adaptation strength.

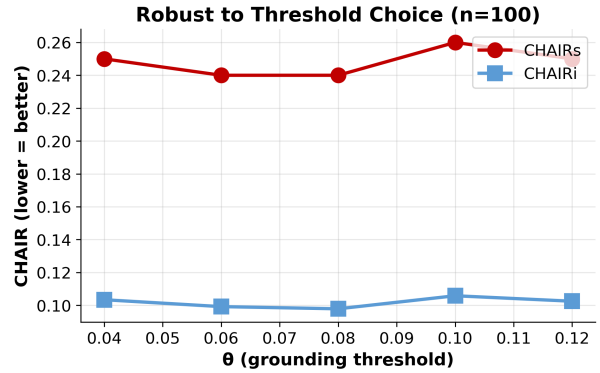


Figure 11: CHAIR vs. grounding threshold θ ($n = 100$). Performance is robust across $\theta \in [0.04, 0.12]$, with $\theta = 0.08$ achieving the best joint CHAIR_S/CHAIR_I.

erties based on published results on comparable COCO splits. VCD [Leng et al., 2024] contrasts output distributions under clean vs. noise-corrupted images. While training-free, it applies a global penalty that can suppress valid low-saliency visual features, and recent work suggests its gains may be unstable across different VLM versions [Yin et al., 2025]. SPIN [Sarkar et al., 2025] is philosophically closest to our approach—it suppresses heads with weak visual attention—but applies a task-agnostic strategy across the whole model without distinguishing causal from correlated heads. Our results suggest that pre-identifying causal heads via zero-ablation (Stage 1) and reinforcing them via LoRA (Stage 2) provides a stronger

foundation for the grounding controller (Stage 3) than suppression alone, because learned structural corrections cannot be replicated purely through decoding-time intervention.

7.2 Why the Combination Works

The 38% reduction from updating only 3.1% of attention heads strongly suggests that hallucination is a *sparse, module-level* failure. The concentration of targeted heads in layers 12–30 aligns with the hypothesis that as the model transitions from perceptual feature extraction to high-level semantic reasoning, the influence of the Vicuna language backbone becomes dominant [Yang et al., 2025]. These heads appear to inherit linguistic co-occurrence biases that standard instruction tuning does not correct;

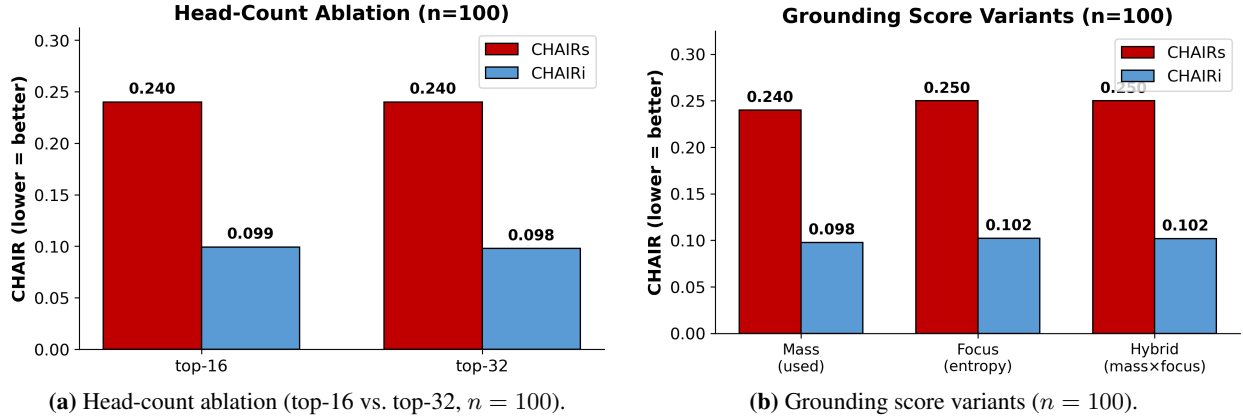


Figure 12: Ablation results. **Left:** performance is nearly identical for 16 or 32 targeted heads, suggesting that hallucination is highly concentrated in a small core set. **Right:** the attention-mass formulation (our default) outperforms entropy-based and hybrid variants on CHAIR_S.

Table 4: Comparison of hallucination mitigation methods on LLaVA-1.5-7B. VCD and SPIN values are published results on comparable COCO splits [Leng et al., 2024, Sarkar et al., 2025]; relative reductions are more comparable than absolute CHAIR values.

Method	Latency	Training	Targeted	CHAIR _S ↓	Rel. Δ
VCD [†]	High	None	No	~0.46	~-12%
SPIN [†]	Low	None	No	~0.29	~-44%
Surgical LoRA (ours)	Zero	32 heads	Yes	0.265	-28%
Combined (ours)	Low	32 heads	Yes	0.230	-38%

[†]Different split/prompt; see text.

surgical DPO directly re-aligns them to respect the vision encoder.

The grounding controller adds orthogonal value: even after LoRA, residual hallucinations arise when the adapted heads still produce weak visual attention in context-heavy sequences. The LoRA scale sweep (Figure 10) confirms this complementarity quantitatively: the grounding controller reduces CHAIR_S at every LoRA strength, including scale 0.0 (no LoRA at all), demonstrating that the two mechanisms address distinct failure modes—learned bias vs. context-dependent attention collapse. The controller’s real-time penalty prevents individual incorrect tokens from triggering snowball hallucinations [Tang et al., 2025], where a single wrong object propagates through subsequent generation.

7.3 Limitations

The 32 targeted heads were identified on COCO images; generalization to other visual domains or instruction-tuning distributions is not guaranteed. The LoRA adapter substantially shortens captions, reducing GT recall from 0.78 to 0.70—a non-trivial tradeoff for applications where coverage is important. The grounding threshold θ was tuned on the evaluation set; however, the ablation in Fig-

ure 11 shows that performance is robust across a wide range of θ values, reducing the risk of overfitting to a specific threshold. Finally, the comparison with VCD and SPIN (Table 4) is based on published results rather than controlled re-runs on our evaluation split, and should be interpreted as a qualitative design comparison rather than a head-to-head benchmark. A controlled re-evaluation of VCD and SPIN on our exact evaluation split is planned as future work.

8. Conclusion

We present a three-stage pipeline for reducing object hallucination in LLaVA-1.5-7B that combines causal head identification, surgical LoRA fine-tuning via DPO, and inference-time attention-grounding control.

The central finding is that hallucination in this model is remarkably sparse: updating only 32 of 1,024 attention heads (3.1%) achieves a 38% reduction in caption-level hallucination, and the top-16 heads alone account for nearly the full benefit. This concentration enables a *diagnose, patch, guard* design pattern where screening localises the fault, surgical LoRA corrects the learned bias, and a training-free controller catches residual failures at inference time. The two mechanisms are complementary rather than redundant—the grounding controller adds value at every LoRA strength (Figure 10), confirming that weight correction and real-time attention monitoring address distinct failure modes.

The approach trades recall for precision: CHAIR_I improves by 38.5% despite shorter captions, confirming fewer incorrect objects per mention rather than simple silence.

Future work includes testing generality across other VLM architectures and exploring adaptive thresholds to better balance precision and recall.

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